

Algebraic Fractions

Worked Solutions with TikZ Diagrams

Wisest Maths — Year 1 A-Level, Algebra (ref a3)

This document contains all 52 *Algebraic Fractions* questions, each with a fully worked solution and a TikZ diagram chosen to match the operation:

- **Cancellation** diagrams (factorise, then strike the common factor) for *simplifying* a single fraction;
- **Cross-multiplication** (butterfly) diagrams for *adding or subtracting* two fractions;
- **Solution-flow** diagrams (one operation per arrow) for *multiplying/dividing* and for common-denominator work with three or more terms.

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Algebraic Fractions 1 (a3-001)

Difficulty: Foundation / Marks: 2

Simplify the expression: $\frac{2x+10}{6}$

Worked solution.

Step 1: Factorise the numerator by extracting the highest common factor of 2.

$$\frac{2(x+5)}{6}$$

Identify the common factor in the top expression to prepare for cancellation.

Step 2: Cancel the common factor of 2 from both the numerator and the denominator.

$$\frac{x+5}{3}$$

Divide the top and bottom by 2.

Final answer: $\frac{x+5}{3}$

Diagram.

$$\frac{\cancel{2}(x+5)}{\cancel{2} \cdot 3} \xrightarrow{\text{cancel } 2} = \frac{x+5}{3}$$

Cancellation diagram: factorise top and bottom, then strike the common factor.

Algebraic Fractions 2 (a3-002)

Difficulty: Foundation / Marks: 2

Simplify fully: $\frac{np^2-2n^2p}{np}$

Worked solution.

Step 1: Factorise the numerator completely by taking out the common factor of np.

$$\frac{np(p-2n)}{np}$$

Find the highest common algebraic factors in the top expression.

Step 2: Cancel the np term from both the numerator and the denominator.

$$p-2n$$

Divide the expression by the common term.

Final answer: $p-2n$

Diagram.

$$\frac{\cancel{np}(p-2n)}{\cancel{np}} \xrightarrow{\text{cancel } np} = p-2n$$

Cancellation diagram: factorise top and bottom, then strike the common factor.

Algebraic Fractions 3 (a3-003)

Difficulty: Foundation / Marks: 2

Express as a single fraction: $\frac{x}{3} + \frac{x}{4}$

Worked solution.

Step 1: Find a common denominator for the fractions, which is 12. Multiply the top and bottom of the first fraction by 4, and the second by 3.

$$\frac{4x}{12} + \frac{3x}{12}$$

Rewrite the fractions so they can be combined.

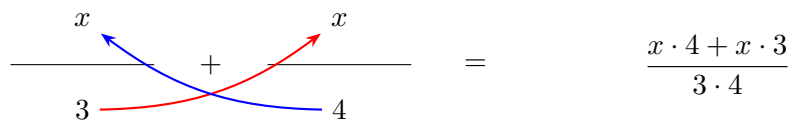
Step 2: Add the numerators together over the common denominator.

$$\frac{7x}{12}$$

Combine the like terms on the top line.

Final answer: $\frac{7x}{12}$

Diagram.

$$\frac{x}{3} + \frac{x}{4} = \frac{x \cdot 4 + x \cdot 3}{3 \cdot 4}$$


Cross-multiplication: each numerator multiplies the *other* denominator (curved arrows); the denominators multiply to give the common denominator.

Algebraic Fractions 4 (a3-004)

Difficulty: Foundation / Marks: 2

Express as a single fraction: $\frac{1}{2p} - \frac{1}{5q}$

Worked solution.

Step 1: Find a common denominator by multiplying the two denominators together to get $10pq$.

$$\frac{5q}{10pq} - \frac{2p}{10pq}$$

Multiply the top and bottom of each fraction by the missing factors.

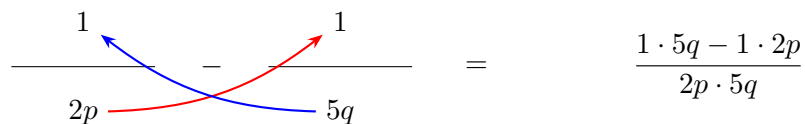
Step 2: Subtract the numerators over the common denominator.

$$\frac{5q - 2p}{10pq}$$

Combine the terms on the top line into a single expression.

Final answer: $\frac{5q - 2p}{10pq}$

Diagram.

$$\frac{1}{2p} - \frac{1}{5q} = \frac{1 \cdot 5q - 1 \cdot 2p}{2p \cdot 5q}$$


Cross-multiplication: each numerator multiplies the *other* denominator (curved arrows); the denominators multiply to give the common denominator.

Algebraic Fractions 5 (a3-005)

Difficulty: Foundation / Marks: 3

Express as a single fraction in its simplest form: $\frac{5}{y-1} + \frac{3}{y-2}$

Worked solution.

Step 1: Find the common denominator by multiplying the two denominators together: $(y - 1)(y - 2)$. Rewrite each fraction.

$$\frac{5(y - 2)}{(y - 1)(y - 2)} + \frac{3(y - 1)}{(y - 1)(y - 2)}$$

Multiply the top and bottom of each fraction by the opposite denominator.

Step 2: Combine the numerators over the common denominator and expand the brackets.

$$\frac{5y - 10 + 3y - 3}{(y - 1)(y - 2)}$$

Bring the fractions together and multiply out the terms on top.

Step 3: Simplify the numerator by collecting like terms: $5y + 3y = 8y$ and $-10 - 3 = -13$.

$$\frac{8y - 13}{(y - 1)(y - 2)}$$

Neaten the top line to get the final answer.

Final answer: $\frac{8y - 13}{(y - 1)(y - 2)}$

Diagram.

$$\frac{5}{(y - 1)} + \frac{3}{(y - 2)} = \frac{5 \cdot (y - 2) + 3 \cdot (y - 1)}{(y - 1) \cdot (y - 2)}$$

Cross-multiplication: each numerator multiplies the *other* denominator (curved arrows); the denominators multiply to give the common denominator.

Algebraic Fractions 6 (a3-006)

Difficulty: Foundation / Marks: 3

Express as a single fraction in its simplest form: $\frac{7}{r-5} - \frac{4}{r+3}$

Worked solution.

Step 1: Find the common denominator: $(r - 5)(r + 3)$. Multiply the numerator of each fraction by the denominator of the other.

$$\frac{7(r + 3)}{(r - 5)(r + 3)} - \frac{4(r - 5)}{(r - 5)(r + 3)}$$

Create equivalent fractions with the same denominator.

Step 2: Combine the numerators and expand the brackets. Pay close attention to expanding the negative term: $-4(r - 5) = -4r + 20$.

$$\frac{7r + 21 - 4r + 20}{(r - 5)(r + 3)}$$

Distribute the values across the brackets on the top line.

Step 3: Collect like terms on the numerator to fully simplify the expression.

$$\frac{3r + 41}{(r - 5)(r + 3)}$$

Combine the r terms and the number terms.

Final answer: $\frac{3r + 41}{(r - 5)(r + 3)}$

Diagram.

$$\frac{\begin{array}{cc} 7 & 4 \\ \swarrow & \searrow \\ (r-5) & (r+3) \end{array}}{(r-5)(r+3)} = \frac{7 \cdot (r+3) - 4 \cdot (r-5)}{(r-5) \cdot (r+3)}$$

Cross-multiplication: each numerator multiplies the *other* denominator (curved arrows); the denominators multiply to give the common denominator.

Algebraic Fractions 7 (a3-007)

Difficulty: Foundation / Marks: 4

Simplify the expression fully: $\frac{4g^2 - 4h^2}{g^2 + gh}$

Worked solution.

Step 1: Factorise the numerator by extracting the common factor of 4.

$$\frac{4(g^2 - h^2)}{g^2 + gh}$$

Always look for the highest common numerical factor first.

Step 2: Notice the bracket on the top is a difference of two squares. Factorise it fully. Also factorise the denominator by extracting g.

$$\frac{4(g - h)(g + h)}{g(g + h)}$$

Break down both the top and bottom expressions into their factored forms.

Step 3: Cancel the common binomial factor of (g + h) from the numerator and denominator.

$$\frac{4(g - h)}{g}$$

Divide the top and bottom by the shared bracket.

Final answer: $\frac{4(g - h)}{g}$

Diagram.

$$\frac{4(g - h)\cancel{(g + h)}}{g\cancel{(g + h)}} \xrightarrow{\text{cancel } (g + h)} = \frac{4(g - h)}{g}$$

Cancellation diagram: factorise top and bottom, then strike the common factor.

Algebraic Fractions 8 (a3-008)

Difficulty: Foundation / Marks: 2

Simplify the expression fully: $\frac{15m-10}{5}$

Worked solution.

Step 1: Factorise the numerator by extracting the highest common factor, which is 5.

$$\frac{5(3m-2)}{5}$$

Look for the largest number that divides exactly into both 15m and 10.

Step 2: Cancel the common factor of 5 from the numerator and the denominator.

$$3m-2$$

Dividing the top and bottom by 5 leaves the expression in the bracket.

Final answer: $3m-2$

Diagram.

$$\frac{\cancel{5}(3m-2)}{\cancel{5}} \xrightarrow{\text{cancel } 5} = 3m-2$$

Cancellation diagram: factorise top and bottom, then strike the common factor.

Algebraic Fractions 9 (a3-009)

Difficulty: Foundation / Marks: 3

Simplify fully: $\frac{x^2+7x+10}{x+5}$

Worked solution.

Step 1: Factorise the quadratic expression in the numerator.

$$\frac{(x+2)(x+5)}{x+5}$$

Find two numbers that multiply to make 10 and add to make 7. These are 2 and 5.

Step 2: Cancel the common binomial factor of $(x+5)$.

$$x+2$$

Divide both the top and bottom by the shared bracket.

Final answer: $x+2$

Diagram.

$$\frac{\cancel{(x+5)}(x+2)}{\cancel{(x+5)}} \xrightarrow{\text{cancel } (x+5)} = x+2$$

Cancellation diagram: factorise top and bottom, then strike the common factor.

Algebraic Fractions 10 (a3-010)

Difficulty: Foundation / Marks: 2

Simplify the expression: $\frac{y^2-9}{y+3}$

Worked solution.

Step 1: Recognise the numerator as a difference of two squares and factorise it.

$$\frac{(y-3)(y+3)}{y+3}$$

y^2 and 9 are both square terms separated by a minus sign.

Step 2: Cancel the common factor $(y+3)$.

$$y-3$$

The bracket $(y+3)$ appears on both the top and bottom, so they cancel out.

Final answer: $y-3$

Diagram.

$$\frac{(y-3)\cancel{(y+3)}}{\cancel{(y+3)}} \xrightarrow{\text{cancel } (y+3)} = y-3$$

Cancellation diagram: factorise top and bottom, then strike the common factor.

Algebraic Fractions 11 (a3-011)

Difficulty: Foundation / Marks: 3

Simplify fully: $\frac{8p^3q}{2pq^2}$

Worked solution.

Step 1: Divide the numerical coefficients.

$$\frac{4p^3q}{pq^2}$$

8 divided by 2 is 4.

Step 2: Use the laws of indices to simplify the p terms.

$$\frac{4p^2q}{q^2}$$

$$p^3 \div p^1 = p^{3-1} = p^2.$$

Step 3: Use the laws of indices to simplify the q terms.

$$\frac{4p^2}{q}$$

$q^1 \div q^2 = q^{-1}$, which places a single q in the denominator.

Final answer: $\frac{4p^2}{q}$

Diagram.

$$\frac{\cancel{2pq} \cdot 4p^2}{\cancel{2pq} \cdot q} \xrightarrow{\text{cancel } 2pq} = \frac{4p^2}{q}$$

Cancellation diagram: factorise top and bottom, then strike the common factor.

Algebraic Fractions 12 (a3-012)

Difficulty: Foundation / Marks: 2

Write as a single fraction in its simplest form: $\frac{3x}{4} + \frac{x}{4}$

Worked solution.

Step 1: Add the numerators together since the denominators are the same.

$$\frac{3x + x}{4}$$

When fractions have a common denominator, you can combine the top terms over the single denominator.

Step 2: Collect like terms on the numerator.

$$\frac{4x}{4}$$

$$3x + 1x = 4x.$$

Step 3: Simplify the fraction by cancelling the common factor of 4.

$$x$$

4 divided by 4 is 1, leaving just x .

Final answer: x

Diagram.

$$\frac{3x}{4} + \frac{x}{4} \xrightarrow{\text{same denominator}} \frac{3x + x}{4} = \frac{4x}{4} \xrightarrow{\text{cancel } 4} x$$

Solution flow: each arrow shows one operation taking the expression to the next form.

Algebraic Fractions 13 (a3-013)

Difficulty: Foundation / Marks: 2

Express as a single fraction: $\frac{x}{2} + \frac{x}{3}$

Worked solution.

Step 1: Find a common denominator for 2 and 3, which is 6.

$$\frac{3x}{6} + \frac{2x}{6}$$

Multiply the top and bottom of the first fraction by 3, and the top and bottom of the second fraction by 2.

Step 2: Add the numerators over the common denominator.

$$\frac{3x + 2x}{6}$$

Combine the terms now that the denominators match.

Step 3: Simplify the numerator.

$$\frac{5x}{6}$$

Collect the like terms: $3x + 2x = 5x$.

Final answer: $\frac{5x}{6}$

Diagram.

$$\frac{x}{2} + \frac{x}{3} = \frac{x \cdot 3 + x \cdot 2}{2 \cdot 3}$$

Cross-multiplication: each numerator multiplies the *other* denominator (curved arrows); the denominators multiply to give the common denominator.

Algebraic Fractions 14 (a3-014)

Difficulty: Foundation / Marks: 3

Simplify fully: $\frac{2x^2+6x}{x^2+3x}$

Worked solution.

Step 1: Factorise the numerator by taking out the common factor of $2x$.

$$\frac{2x(x+3)}{x^2+3x}$$

$2x$ is the highest common factor of both $2x^2$ and $6x$.

Step 2: Factorise the denominator by taking out the common factor of x .

$$\frac{2x(x+3)}{x(x+3)}$$

x is the highest common factor of both x^2 and $3x$.

Step 3: Cancel the common factors x and $(x+3)$ from the top and bottom.

$$2$$

Both the numerator and denominator share the algebraic factor $x(x+3)$.

Final answer: 2

Diagram.

$$\frac{\cancel{2x(x+3)}}{\cancel{x(x+3)}} \xrightarrow{\text{cancel } x(x+3)} = 2$$

Cancellation diagram: factorise top and bottom, then strike the common factor.

Algebraic Fractions 15 (a3-015)

Difficulty: Foundation / Marks: 2

Simplify fully: $\frac{2x}{3} \times \frac{5}{4x}$

Worked solution.

Step 1: Multiply the numerators together and the denominators together.

$$\frac{2x \times 5}{3 \times 4x}$$

When multiplying fractions, you multiply straight across the top and straight across the bottom.

Step 2: Simplify the resulting fraction.

$$\frac{10x}{12x}$$

Combine the terms to get a single numerator and denominator.

Step 3: Cancel the common factors from the top and bottom.

$$\frac{5}{6}$$

Divide both the top and bottom by 2 and cancel out the x .

Final answer: $\frac{5}{6}$

Diagram.

$$\frac{2x}{3} \times \frac{5}{4x} \xrightarrow{\text{multiply across}} \frac{2x \cdot 5}{3 \cdot 4x} = \frac{10x}{12x} \xrightarrow{\text{cancel } 2x} \frac{5}{6}$$

Solution flow: each arrow shows one operation taking the expression to the next form.

Algebraic Fractions 16 (a3-016)

Difficulty: Foundation / Marks: 3

Simplify fully: $\frac{3a}{b} \div \frac{6a^2}{b^2}$

Worked solution.

Step 1: Change the division into a multiplication by flipping the second fraction.

$$\frac{3a}{b} \times \frac{b^2}{6a^2}$$

Use the rule: Keep the first fraction, Change the sign to multiply, Flip the second fraction.

Step 2: Multiply the numerators and the denominators.

$$\frac{3ab^2}{6a^2b}$$

Combine into a single algebraic fraction.

Step 3: Simplify the coefficients and use index laws to cancel the variables.

$$\frac{b}{2a}$$

$\frac{3}{6}$ simplifies to $\frac{1}{2}$. $\frac{a}{a^2}$ leaves a on the bottom. $\frac{b^2}{b}$ leaves b on the top.

Final answer: $\frac{b}{2a}$

Diagram.

$$\frac{3a}{b} \div \frac{6a^2}{b^2} \xrightarrow{\text{keep-change-flip}} \frac{3a}{b} \times \frac{b^2}{6a^2} \xrightarrow{\text{multiply across}} \frac{3a b^2}{6a^2 b} \xrightarrow{\text{cancel } 3ab} \frac{b}{2a}$$

Solution flow: each arrow shows one operation taking the expression to the next form.

Algebraic Fractions 17 (a3-017)

Difficulty: Foundation / Marks: 3

Express as a single fraction in its simplest form: $\frac{5}{x+1} - \frac{2}{x-2}$

Worked solution.

Step 1: Find a common denominator by multiplying the two denominators together.

$$\frac{5(x-2)}{(x+1)(x-2)} - \frac{2(x+1)}{(x+1)(x-2)}$$

Multiply the top and bottom of each fraction by the denominator of the other fraction.

Step 2: Combine the numerators over the single common denominator.

$$\frac{5(x-2) - 2(x+1)}{(x+1)(x-2)}$$

Now that the denominators are the same, subtract the second numerator from the first.

Step 3: Expand the brackets in the numerator and collect like terms.

$$\frac{5x - 10 - 2x - 2}{(x+1)(x-2)}$$

Be careful with the negative sign when expanding $-2(x+1)$.

Step 4: Simplify the numerator to get the final answer.

$$\frac{3x - 12}{(x+1)(x-2)}$$

$5x - 2x = 3x$ and $-10 - 2 = -12$.

Final answer: $\frac{3x - 12}{(x + 1)(x - 2)}$

Diagram.

$$\frac{5}{(x+1)} - \frac{2}{(x-2)} = \frac{5 \cdot (x-2) - 2 \cdot (x+1)}{(x+1) \cdot (x-2)}$$

Cross-multiplication: each numerator multiplies the *other* denominator (curved arrows); the denominators multiply to give the common denominator.

Algebraic Fractions 18 (a3-018)

Difficulty: Foundation / Marks: 4

Simplify fully: $\frac{x^2 - x - 12}{x^2 - 16}$

Worked solution.

Step 1: Factorise the quadratic expression in the numerator.

$$\frac{(x - 4)(x + 3)}{x^2 - 16}$$

Find numbers that multiply to -12 and add to -1. These are -4 and 3.

Step 2: Factorise the denominator, which is a difference of two squares.

$$\frac{(x - 4)(x + 3)}{(x - 4)(x + 4)}$$

x^2 and 16 are both square terms, so it factorises to $(x - 4)(x + 4)$.

Step 3: Cancel the common factor of $(x - 4)$ from the numerator and denominator.

$$\frac{x + 3}{x + 4}$$

Divide the top and bottom by their shared bracket.

Final answer: $\frac{x + 3}{x + 4}$

Diagram.

$$\frac{\cancel{(x-4)}(x+3)}{\cancel{(x-4)}(x+4)} \xrightarrow{\text{cancel } (x-4)} = \frac{x+3}{x+4}$$

Cancellation diagram: factorise top and bottom, then strike the common factor.

Algebraic Fractions 19 (a3-019)

Difficulty: Foundation / Marks: 2

Write as a single fraction: $\frac{2}{x} + \frac{3}{2x}$

Worked solution.

Step 1: Identify the lowest common denominator, which is $2x$. Convert the first fraction.

$$\frac{4}{2x} + \frac{3}{2x}$$

Multiply the top and bottom of the first fraction by 2 so both fractions share the denominator $2x$.

Step 2: Add the numerators together.

$$\frac{4+3}{2x}$$

Since the denominators match, you can simply add the top values.

Step 3: Simplify to get the final fraction.

$$\frac{7}{2x}$$

Evaluate $4 + 3$.

Final answer: $\frac{7}{2x}$

Diagram.

$$\frac{2}{x} + \frac{3}{2x} \xrightarrow{\text{3 common denominator}} \frac{2x}{2x} + \frac{3}{2x} \xrightarrow{\text{add numerators}} \frac{7}{2x}$$

Solution flow: each arrow shows one operation taking the expression to the next form.

Algebraic Fractions 20 (a3-020)

Difficulty: Foundation / Marks: 3

Simplify fully: $\frac{x+2}{3} \times \frac{9}{x^2-4}$

Worked solution.

Step 1: Factorise the expression $x^2 - 4$ in the denominator.

$$\frac{x+2}{3} \times \frac{9}{(x-2)(x+2)}$$

Recognise this as a difference of two squares.

Step 2: Cross-cancel the common numerical factors.

$$\frac{x+2}{1} \times \frac{3}{(x-2)(x+2)}$$

Divide the 9 on the top right and the 3 on the bottom left by 3.

Step 3: Cross-cancel the common algebraic factor $(x+2)$.

$$1 \times \frac{3}{x-2}$$

The $(x + 2)$ on the top left cancels with the $(x + 2)$ on the bottom right.

Step 4: Multiply the remaining parts together.

$$\frac{3}{x - 2}$$

Combine what is left after cancelling.

Final answer: $\frac{3}{x - 2}$

Diagram.

$$\frac{x + 2}{3} \times \frac{9}{x^2 - 4} \xrightarrow{\text{factorise } x^2 - 4} \frac{x + 2}{3} \times \frac{9}{(x - 2)(x + 2)} \xrightarrow{\text{cancel } (x + 2) \text{ and } 9/3} \frac{3}{x - 2}$$

Solution flow: each arrow shows one operation taking the expression to the next form.

Algebraic Fractions 21 (a3-021)

Difficulty: Foundation / Marks: 3

Simplify fully: $\frac{5-x}{x^2-25}$

Worked solution.

Step 1: Factorise the denominator as a difference of two squares.

$$\frac{5 - x}{(x - 5)(x + 5)}$$

$x^2 - 25$ factorises to $(x - 5)(x + 5)$.

Step 2: Factorise out a -1 from the numerator to reverse the terms.

$$\frac{-(x - 5)}{(x - 5)(x + 5)}$$

$5 - x$ is the exact negative of $x - 5$, so pulling out a -1 creates a matching bracket.

Step 3: Cancel the matching $(x - 5)$ bracket.

$$\frac{-1}{x + 5}$$

Divide the top and bottom by $(x - 5)$.

Final answer: $-\frac{1}{x + 5}$

Diagram.

$$\frac{-(\cancel{x - 5})}{(\cancel{x - 5})(x + 5)} \xrightarrow{\text{cancel } (x - 5)} = -\frac{1}{x + 5}$$

Cancellation diagram: factorise top and bottom, then strike the common factor.

Algebraic Fractions 22 (a3-022)

Difficulty: Foundation / Marks: 3

Express as a single fraction: $\frac{2x-1}{3} + \frac{x+4}{5}$

Worked solution.

Step 1: Find a common denominator, which is 15.

$$\frac{5(2x-1)}{15} + \frac{3(x+4)}{15}$$

Multiply the top and bottom of the first fraction by 5, and the second by 3.

Step 2: Expand the brackets on the numerators and write over a single fraction.

$$\frac{10x - 5 + 3x + 12}{15}$$

Multiply the terms outside the brackets by the terms inside.

Step 3: Collect like terms on the numerator to simplify fully.

$$\frac{13x + 7}{15}$$

$10x + 3x = 13x$, and $-5 + 12 = 7$.

Final answer: $\frac{13x + 7}{15}$

Diagram.

$$\frac{(2x-1)}{3} + \frac{(x+4)}{5} = \frac{(2x-1) \cdot 5 + (x+4) \cdot 3}{3 \cdot 5}$$

Cross-multiplication: each numerator multiplies the *other* denominator (curved arrows); the denominators multiply to give the common denominator.

Algebraic Fractions 23 (a3-023)

Difficulty: Foundation / Marks: 2

Simplify fully: $\frac{2x+10}{6}$

Worked solution.

Step 1: Factorise the numerator by taking out the highest common factor of 2.

$$\frac{2(x+5)}{6}$$

Both $2x$ and 10 divide exactly by 2 .

Step 2: Cancel the common factor from the top and bottom.

$$\frac{x+5}{3}$$

Divide the 2 on the top and the 6 on the bottom by 2 .

Final answer: $\frac{x+5}{3}$

Diagram.

$$\frac{\cancel{2}(x+5)}{\cancel{2} \cdot 3} \xrightarrow{\text{cancel 2}} = \frac{x+5}{3}$$

Cancellation diagram: factorise top and bottom, then strike the common factor.

Algebraic Fractions 24 (a3-024)

Difficulty: Foundation / Marks: 2

Simplify fully: $\frac{6a-12b-15c}{3}$

Worked solution.

Step 1: Factorise the expression on the top line by finding the highest common numerical factor, which is 3.

$$\frac{3(2a - 4b - 5c)}{3}$$

3 divides exactly into 6a, -12b, and -15c.

Step 2: Cancel the common factor of 3 from the numerator and denominator.

$$2a - 4b - 5c$$

Dividing the top and bottom by 3 removes the fractional part entirely.

Final answer: $2a - 4b - 5c$

Diagram.

$$\frac{\cancel{3}(2a - 4b - 5c)}{\cancel{3}} \xrightarrow{\text{cancel 3}} = 2a - 4b - 5c$$

Cancellation diagram: factorise top and bottom, then strike the common factor.

Algebraic Fractions 25 (a3-025)

Difficulty: Foundation / Marks: 3

Simplify fully: $\frac{np^2-2n^2p}{np}$

Worked solution.

Step 1: Factorise the numerator by extracting the highest common algebraic factor, which is np.

$$\frac{np(p - 2n)}{np}$$

Both terms in the numerator share an n and a p.

Step 2: Cancel the shared factor of np from the top and bottom.

$$p - 2n$$

Dividing the top and bottom by np leaves just the bracketed expression.

Final answer: $p - 2n$

Diagram.

$$\frac{\cancel{np}(p-2n)}{\cancel{np}} \xrightarrow{\text{cancel } np} = p - 2n$$

Cancellation diagram: factorise top and bottom, then strike the common factor.

Algebraic Fractions 26 (a3-026)

Difficulty: Foundation / Marks: 3

Express as a single fraction in its simplest form: $\frac{5}{y-1} + \frac{3}{y-2}$

Worked solution.

Step 1: Find a common denominator by multiplying the two linear brackets together.

$$\frac{5(y-2)}{(y-1)(y-2)} + \frac{3(y-1)}{(y-1)(y-2)}$$

Multiply the top and bottom of each fraction by the denominator of the opposing fraction.

Step 2: Add the numerators together over the common denominator.

$$\frac{5(y-2) + 3(y-1)}{(y-1)(y-2)}$$

Combine the terms now that the denominators match.

Step 3: Expand the brackets on the numerator.

$$\frac{5y - 10 + 3y - 3}{(y-1)(y-2)}$$

Multiply the numbers outside the brackets by the terms inside.

Step 4: Collect like terms to fully simplify the numerator.

$$\frac{8y - 13}{(y-1)(y-2)}$$

$5y + 3y = 8y$ and $-10 - 3 = -13$.

Final answer: $\frac{8y - 13}{(y-1)(y-2)}$

Diagram.

$$\begin{array}{c} 5 \qquad \qquad \qquad 3 \\ \swarrow \quad \quad \searrow \\ \hline (y-1) \quad \quad (y-2) \end{array} + = \frac{5 \cdot (y-2) + 3 \cdot (y-1)}{(y-1) \cdot (y-2)}$$

Cross-multiplication: each numerator multiplies the *other* denominator (curved arrows); the denominators multiply to give the common denominator.

Algebraic Fractions 27 (a3-027)

Difficulty: Foundation / Marks: 3

Express as a single fraction in its simplest form: $\frac{7}{r-5} - \frac{4}{r+3}$

Worked solution.

Step 1: Find a common denominator by multiplying the two denominators.

$$\frac{7(r+3)}{(r-5)(r+3)} - \frac{4(r-5)}{(r-5)(r+3)}$$

Cross-multiply the numerators and denominators to find equivalent fractions.

Step 2: Combine the numerators over the single denominator.

$$\frac{7(r+3) - 4(r-5)}{(r-5)(r+3)}$$

Subtract the second numerator from the first.

Step 3: Expand the brackets on the numerator carefully, paying attention to the negative sign.

$$\frac{7r + 21 - 4r + 20}{(r-5)(r+3)}$$

Be careful to note that $-4 \times -5 = +20$.

Step 4: Collect like terms on the numerator to finish.

$$\frac{3r + 41}{(r-5)(r+3)}$$

$7r - 4r = 3r$ and $21 + 20 = 41$.

Final answer: $\frac{3r + 41}{(r-5)(r+3)}$

Diagram.

$$\frac{7}{(r-5)} - \frac{4}{(r+3)} = \frac{7 \cdot (r+3) - 4 \cdot (r-5)}{(r-5) \cdot (r+3)}$$

Cross-multiplication: each numerator multiplies the *other* denominator (curved arrows); the denominators multiply to give the common denominator.

Algebraic Fractions 28 (a3-028)

Difficulty: Foundation / Marks: 4

Express as a single fraction in its simplest form: $\frac{z+1}{z+2} - \frac{z+3}{z+4}$

Worked solution.

Step 1: Form equivalent fractions with a common denominator of $(z+2)(z+4)$.

$$\frac{(z+1)(z+4)}{(z+2)(z+4)} - \frac{(z+3)(z+2)}{(z+2)(z+4)}$$

Multiply the top and bottom of each fraction by the opposing denominator.

Step 2: Write as a single fraction.

$$\frac{(z+1)(z+4) - (z+3)(z+2)}{(z+2)(z+4)}$$

Combine the numerators over the shared denominator.

Step 3: Expand the double brackets on the numerator.

$$\frac{(z^2 + 5z + 4) - (z^2 + 5z + 6)}{(z+2)(z+4)}$$

Multiply out each pair of brackets completely.

Step 4: Subtract the expanded terms and simplify the numerator.

$$\frac{-2}{(z+2)(z+4)}$$

$z^2 - z^2 = 0$, $5z - 5z = 0$, and $4 - 6 = -2$.

Final answer: $\frac{-2}{(z+2)(z+4)}$

Diagram.

$$\frac{(z+1)}{(z+2)} - \frac{(z+3)}{(z+4)} = \frac{(z+1) \cdot (z+4) - (z+3) \cdot (z+2)}{(z+2) \cdot (z+4)}$$

Cross-multiplication: each numerator multiplies the *other* denominator (curved arrows); the denominators multiply to give the common denominator.

Algebraic Fractions 29 (a3-029)

Difficulty: Foundation / Marks: 2

Express as a single fraction: $\frac{2}{t} + \frac{13}{r^2}$

Worked solution.

Step 1: Identify the common denominator by multiplying the two distinct denominators together.

$$\frac{2(r^2)}{tr^2} + \frac{13(t)}{tr^2}$$

Multiply the first fraction by r^2 on the top and bottom, and the second fraction by t on the top and bottom.

Step 2: Combine the numerators over the single common denominator.

$$\frac{2r^2 + 13t}{tr^2}$$

Since the variables are different, you cannot collect the terms any further.

Final answer: $\frac{2r^2 + 13t}{tr^2}$

Diagram.

$$\frac{2}{t} + \frac{13}{r^2} = \frac{2 \cdot r^2 + 13 \cdot t}{t \cdot r^2}$$

Cross-multiplication: each numerator multiplies the *other* denominator (curved arrows); the denominators multiply to give the common denominator.

Algebraic Fractions 30 (a3-030)

Difficulty: Foundation / Marks: 3

Express as a single fraction in its simplest form: $\frac{8}{p} - \frac{1}{p-3}$

Worked solution.

Step 1: Find a common denominator, which is $p(p-3)$.

$$\frac{8(p-3)}{p(p-3)} - \frac{1(p)}{p(p-3)}$$

Multiply the top and bottom of each fraction by the denominator of the other fraction.

Step 2: Combine the numerators over the single denominator.

$$\frac{8(p-3) - p}{p(p-3)}$$

Subtract the second numerator from the first.

Step 3: Expand the bracket on the numerator.

$$\frac{8p - 24 - p}{p(p-3)}$$

Multiply 8 by p and by -3 .

Step 4: Collect like terms to simplify the numerator fully.

$$\frac{7p - 24}{p(p-3)}$$

$$8p - 1p = 7p.$$

Final answer: $\frac{7p - 24}{p(p-3)}$

Diagram.

$$\frac{8}{p} - \frac{1}{p-3} = \frac{8 \cdot (p-3) - 1 \cdot p}{p \cdot (p-3)}$$

Cross-multiplication: each numerator multiplies the *other* denominator (curved arrows); the denominators multiply to give the common denominator.

Algebraic Fractions 31 (a3-031)

Difficulty: Foundation / Marks: 3

Simplify fully: $\frac{10yz-40y^3z+60y^2z^3}{10z^2}$

Worked solution.

Step 1: Factorise the numerator by extracting the highest common factor of all three terms, which is $10yz$.

$$\frac{10yz(1 - 4y^2 + 6yz^2)}{10z^2}$$

Look for the largest number and the highest powers of the variables that divide exactly into every term.

Step 2: Cancel the common factor of $10z$ from the numerator and denominator.

$$\frac{y(1 - 4y^2 + 6yz^2)}{z}$$

Divide the 10 on the top and bottom to get 1. Dividing z by z^2 leaves a single z on the denominator.

Final answer: $\frac{y(1 - 4y^2 + 6yz^2)}{z}$
--

Diagram.

$$\frac{\cancel{10z} \cdot y(1 - 4y^2 + 6yz^2)}{\cancel{10z} \cdot z} \xrightarrow{\text{cancel } 10z} = \frac{y(1 - 4y^2 + 6yz^2)}{z}$$

Cancellation diagram: factorise top and bottom, then strike the common factor.

Algebraic Fractions 32 (a3-032)

Difficulty: Foundation / Marks: 2

Simplify fully: $\frac{4st+6s^2t+9s^3t}{2t}$

Worked solution.

Step 1: Factorise the numerator by extracting the common factor, which is t .

$$\frac{t(4s + 6s^2 + 9s^3)}{2t}$$

There is no common numerical factor other than 1, but every term contains the variable t .

Step 2: Cancel the common factor of t from the top and bottom.

$$\frac{4s + 6s^2 + 9s^3}{2}$$

Dividing the numerator and denominator by t removes it entirely.

Final answer: $\frac{4s + 6s^2 + 9s^3}{2}$

Diagram.

$$\frac{\cancel{t}(4s + 6s^2 + 9s^3)}{\cancel{t} \cdot 2} \xrightarrow{\text{cancel } t} = \frac{4s + 6s^2 + 9s^3}{2}$$

Cancellation diagram: factorise top and bottom, then strike the common factor.

Algebraic Fractions 33 (a3-033)

Difficulty: Foundation / Marks: 3

Express as a single fraction in its simplest form: $\frac{1}{q+1} + \frac{3}{q-2}$

Worked solution.

Step 1: Find a common denominator by multiplying the two linear denominators together.

$$\frac{1(q-2)}{(q+1)(q-2)} + \frac{3(q+1)}{(q+1)(q-2)}$$

Multiply the top and bottom of each fraction by the opposing denominator.

Step 2: Combine the numerators over the single common denominator.

$$\frac{1(q-2) + 3(q+1)}{(q+1)(q-2)}$$

Add the terms together now that the denominators match.

Step 3: Expand the brackets on the numerator.

$$\frac{q-2+3q+3}{(q+1)(q-2)}$$

Multiply the terms outside the brackets by the terms inside.

Step 4: Collect like terms to fully simplify the numerator.

$$\frac{4q+1}{(q+1)(q-2)}$$

$q + 3q = 4q$ and $-2 + 3 = 1$.

Final answer: $\frac{4q+1}{(q+1)(q-2)}$
--

Diagram.

$$\frac{1}{(q+1)} + \frac{3}{(q-2)} = \frac{1 \cdot (q-2) + 3 \cdot (q+1)}{(q+1) \cdot (q-2)}$$

Cross-multiplication: each numerator multiplies the *other* denominator (curved arrows); the denominators multiply to give the common denominator.

Algebraic Fractions 34 (a3-034)

Difficulty: Foundation / Marks: 3

Express as a single fraction in its simplest form: $\frac{x}{x+z} + \frac{2z}{x-z}$

Worked solution.

Step 1: Form equivalent fractions with a common denominator of $(x+z)(x-z)$.

$$\frac{x(x-z)}{(x+z)(x-z)} + \frac{2z(x+z)}{(x+z)(x-z)}$$

Multiply the first fraction by $(x-z)$ top and bottom, and the second by $(x+z)$.

Step 2: Combine the numerators over the shared denominator.

$$\frac{x(x-z) + 2z(x+z)}{(x+z)(x-z)}$$

Add the expanded numerators together.

Step 3: Expand the brackets on the numerator.

$$\frac{x^2 - xz + 2xz + 2z^2}{(x+z)(x-z)}$$

Multiply the outer variables into the brackets.

Step 4: Collect the like terms to simplify the numerator.

$$\frac{x^2 + xz + 2z^2}{(x+z)(x-z)}$$

$$-xz + 2xz = xz.$$

Final answer: $\frac{x^2 + xz + 2z^2}{(x+z)(x-z)}$

Diagram.

$$\frac{x}{(x+z)} + \frac{2z}{(x-z)} = \frac{x \cdot (x-z) + 2z \cdot (x+z)}{(x+z) \cdot (x-z)}$$

Cross-multiplication: each numerator multiplies the *other* denominator (curved arrows); the denominators multiply to give the common denominator.

Algebraic Fractions 35 (a3-035)

Difficulty: Foundation / Marks: 2

Express as a single fraction: $\frac{1}{2p} - \frac{1}{5q}$

Worked solution.

Step 1: Find a common denominator by multiplying the two distinct denominators, which gives $10pq$.

$$\frac{1(5q)}{10pq} - \frac{1(2p)}{10pq}$$

Multiply the first fraction by $5q$ top and bottom, and the second fraction by $2p$.

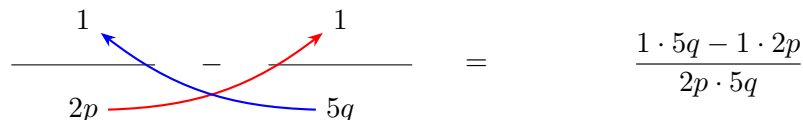
Step 2: Combine the numerators over the single common denominator.

$$\frac{5q - 2p}{10pq}$$

Subtract the numerators. These are not like terms, so they cannot be simplified further.

Final answer: $\frac{5q - 2p}{10pq}$

Diagram.

$$\frac{1}{2p} - \frac{1}{5q} = \frac{1 \cdot 5q - 1 \cdot 2p}{2p \cdot 5q}$$


Cross-multiplication: each numerator multiplies the *other* denominator (curved arrows); the denominators multiply to give the common denominator.

Algebraic Fractions 36 (a3-036)

Difficulty: Foundation / Marks: 3

Express as a single fraction: $\frac{2}{ab^3} - \frac{9}{a^3b}$

Worked solution.

Step 1: Identify the lowest common multiple for the denominators, which is a^3b^3 .

$$\frac{2(a^2)}{ab^3(a^2)} - \frac{9(b^2)}{a^3b(b^2)}$$

To match the denominators, the first fraction needs an extra a^2 and the second fraction needs an extra b^2 .

Step 2: Multiply the terms out.

$$\frac{2a^2}{a^3b^3} - \frac{9b^2}{a^3b^3}$$

Apply the adjustments to create equivalent fractions with matching bases.

Step 3: Combine the numerators over the single common denominator.

$$\frac{2a^2 - 9b^2}{a^3b^3}$$

Subtract the second numerator from the first.

Final answer: $\frac{2a^2 - 9b^2}{a^3b^3}$

Diagram.

$$\frac{2}{ab^3} - \frac{9}{a^3b} \xrightarrow{\text{common denominator } a^3b^3} \frac{2a^2 - 9b^2}{a^3b^3}$$

Solution flow: each arrow shows one operation taking the expression to the next form.

Algebraic Fractions 37 (a3-037)

Difficulty: Foundation / Marks: 4

Express as a single fraction in its simplest form: $\frac{w}{2(w-2)} + \frac{3w}{w-7}$

Worked solution.

Step 1: Find a common denominator, which is $2(w-2)(w-7)$.

$$\frac{w(w-7)}{2(w-2)(w-7)} + \frac{3w(2)(w-2)}{2(w-2)(w-7)}$$

Multiply the top and bottom of the first fraction by $(w-7)$, and the second fraction by $2(w-2)$.

Step 2: Combine the numerators and simplify the multiplier on the second term.

$$\frac{w(w-7) + 6w(w-2)}{2(w-2)(w-7)}$$

Note that $3w \times 2 = 6w$.

Step 3: Expand the brackets on the numerator.

$$\frac{w^2 - 7w + 6w^2 - 12w}{2(w-2)(w-7)}$$

Multiply the terms outside the brackets by the terms inside.

Step 4: Collect like terms to simplify the numerator fully.

$$\frac{7w^2 - 19w}{2(w-2)(w-7)}$$

$w^2 + 6w^2 = 7w^2$ and $-7w - 12w = -19w$.

Final answer: $\frac{7w^2 - 19w}{2(w-2)(w-7)}$

Diagram.

$$\frac{w}{2(w-2)} + \frac{3w}{(w-7)} = \frac{w \cdot (w-7) + 3w \cdot 2(w-2)}{2(w-2) \cdot (w-7)}$$

Cross-multiplication: each numerator multiplies the *other* denominator (curved arrows); the denominators multiply to give the common denominator.

Algebraic Fractions 38 (a3-038)

Difficulty: Foundation / Marks: 3

Simplify fully: $\frac{12cd-6c^2d+3c^3d^2}{12c^2de}$

Worked solution.

Step 1: Factorise the numerator by extracting the highest common factor of all three terms, which is $3cd$.

$$\frac{3cd(4 - 2c + c^2d)}{12c^2de}$$

3 is the largest number that divides into 12, 6, and 3. Each term also contains at least a c and a d .

Step 2: Cancel the common factors of 3, c , and d from the numerator and denominator.

$$\frac{4 - 2c + c^2d}{4ce}$$

Dividing 12 by 3 leaves 4 on the denominator. Dividing c^2 by c leaves c , and the d is removed entirely.

Final answer: $\frac{4 - 2c + c^2d}{4ce}$

Diagram.

$$\frac{\cancel{3cd}(4 - 2c + c^2d)}{\cancel{3cd} \cdot 4ce} \xrightarrow{\text{cancel } 3cd} = \frac{4 - 2c + c^2d}{4ce}$$

Cancellation diagram: factorise top and bottom, then strike the common factor.

Algebraic Fractions 39 (a3-039)

Difficulty: Foundation / Marks: 3

Express as a single fraction: $\frac{2}{mn} - \frac{3m}{n} + \frac{n^2}{m}$

Worked solution.

Step 1: Identify the common denominator for all three fractions, which is mn .

$$\frac{2}{mn} - \frac{3m(m)}{mn} + \frac{n^2(n)}{mn}$$

The first fraction already has the denominator mn . Multiply the top and bottom of the second fraction by m , and the third by n .

Step 2: Simplify the numerators of the equivalent fractions.

$$\frac{2}{mn} - \frac{3m^2}{mn} + \frac{n^3}{mn}$$

Multiply the terms out: $3m \times m = 3m^2$ and $n^2 \times n = n^3$.

Step 3: Combine the numerators over the single common denominator.

$$\frac{2 - 3m^2 + n^3}{mn}$$

Since these are not like terms, they cannot be simplified further.

Final answer: $\frac{2 - 3m^2 + n^3}{mn}$

Diagram.

$$\frac{2}{mn} - \frac{3m}{n} + \frac{n^2}{m} \xrightarrow{\text{common denominator } mn} \frac{2 - 3m^2 + n^3}{mn}$$

Solution flow: each arrow shows one operation taking the expression to the next form.

Algebraic Fractions 40 (a3-040)

Difficulty: Foundation / Marks: 4

Express as a single fraction in its simplest form: $\frac{y}{2x+3} - \frac{2y}{3-x}$

Worked solution.

Step 1: Find a common denominator by multiplying the two linear denominators together.

$$\frac{y(3-x)}{(2x+3)(3-x)} - \frac{2y(2x+3)}{(2x+3)(3-x)}$$

Multiply the top and bottom of each fraction by the opposing denominator.

Step 2: Combine the numerators over the single common denominator.

$$\frac{y(3-x) - 2y(2x+3)}{(2x+3)(3-x)}$$

Subtract the second numerator from the first.

Step 3: Expand the brackets on the numerator.

$$\frac{3y - xy - (4xy + 6y)}{(2x+3)(3-x)}$$

Multiply the terms outside the brackets by the terms inside. Be careful with the negative sign.

Step 4: Collect like terms and simplify fully.

$$\frac{-3y - 5xy}{(2x+3)(3-x)}$$

$3y - 6y = -3y$ and $-xy - 4xy = -5xy$. You can also factorise out a $-y$ on top to get $\frac{-y(3+5x)}{(2x+3)(3-x)}$.

Final answer: $\frac{-3y - 5xy}{(2x+3)(3-x)}$

Diagram.

$$\frac{\begin{array}{cc} y & 2y \\ \swarrow & \searrow \\ (2x+3) & (3-x) \end{array}}{\quad} - \frac{\begin{array}{cc} y & 2y \\ \searrow & \swarrow \\ (2x+3) & (3-x) \end{array}}{\quad} = \frac{y \cdot (3-x) - 2y \cdot (2x+3)}{(2x+3) \cdot (3-x)}$$

Cross-multiplication: each numerator multiplies the *other* denominator (curved arrows); the denominators multiply to give the common denominator.

Algebraic Fractions 41 (a3-041)

Difficulty: Foundation / Marks: 3

Express as a single fraction: $\frac{1}{x} + \frac{2x}{y} + \frac{4}{x^2}$

Worked solution.

Step 1: Identify the lowest common multiple for the denominators, which is x^2y .

$$\frac{1(xy)}{x^2y} + \frac{2x(x^2)}{x^2y} + \frac{4(y)}{x^2y}$$

Adjust each fraction so they all share the denominator x^2y .

Step 2: Simplify the numerators of the equivalent fractions.

$$\frac{xy}{x^2y} + \frac{2x^3}{x^2y} + \frac{4y}{x^2y}$$

Multiply out the terms on the top of each fraction.

Step 3: Combine the numerators over the single common denominator.

$$\frac{xy + 2x^3 + 4y}{x^2y}$$

Add the terms together. They are not like terms, so the expression is complete.

Final answer: $\frac{xy + 2x^3 + 4y}{x^2y}$

Diagram.

$$\frac{1}{x} + \frac{2x}{y} + \frac{4}{x^2} \xrightarrow{\text{common denominator } x^2y} \frac{xy + 2x^3 + 4y}{x^2y}$$

Solution flow: each arrow shows one operation taking the expression to the next form.

Algebraic Fractions 42 (a3-042)

Difficulty: Foundation / Marks: 3

Express as a single fraction: $\frac{ab}{c} + \frac{bc}{a} + \frac{ca}{b}$

Worked solution.

Step 1: Find a common denominator by multiplying all three distinct denominators together to get abc .

$$\frac{ab(ab)}{abc} + \frac{bc(bc)}{abc} + \frac{ca(ca)}{abc}$$

Multiply the top and bottom of each fraction by the variables it is missing from the common denominator.

Step 2: Simplify the numerators by applying the laws of indices.

$$\frac{a^2b^2}{abc} + \frac{b^2c^2}{abc} + \frac{c^2a^2}{abc}$$

For example, $ab \times ab = a^2b^2$.

Step 3: Combine the terms over the single denominator.

$$\frac{a^2b^2 + b^2c^2 + c^2a^2}{abc}$$

Add the numerators together.

Final answer: $\frac{a^2b^2 + b^2c^2 + c^2a^2}{abc}$

Diagram.

$$\frac{ab}{c} + \frac{bc}{a} + \frac{ca}{b} \xrightarrow{\text{common denominator}} \frac{a^2b^2 + b^2c^2 + c^2a^2}{abc}$$

Solution flow: each arrow shows one operation taking the expression to the next form.

Algebraic Fractions 43 (a3-043)

Difficulty: Foundation / Marks: 3

Express as a single fraction: $2 + \frac{a^2}{b} - \frac{2b}{a^3}$

Worked solution.

Step 1: Identify the common denominator, which is a^3b . Convert the whole number 2 into a fraction.

$$\frac{2(a^3b)}{a^3b} + \frac{a^2(a^3)}{a^3b} - \frac{2b(b)}{a^3b}$$

Adjust each term so they share the same denominator.

Step 2: Simplify the numerators of the equivalent fractions.

$$\frac{2a^3b}{a^3b} + \frac{a^5}{a^3b} - \frac{2b^2}{a^3b}$$

Multiply the terms out using index laws where necessary.

Step 3: Combine the numerators over the single common denominator.

$$\frac{2a^3b + a^5 - 2b^2}{a^3b}$$

Add and subtract the terms as written.

Final answer: $\frac{2a^3b + a^5 - 2b^2}{a^3b}$

Diagram.

$$2 + \frac{a^2}{b} - \frac{2b}{a^3} \xrightarrow{\text{common denominator}} \frac{2a^3b + a^5 - 2b^2}{a^3b}$$

Solution flow: each arrow shows one operation taking the expression to the next form.

Algebraic Fractions 44 (a3-044)

Difficulty: Foundation / Marks: 3

Simplify fully: $\frac{2x+x^2y-x^2}{x^2+3x}$

Worked solution.

Step 1: Factorise the numerator by extracting the common factor of x .

$$\frac{x(2 + xy - x)}{x^2 + 3x}$$

Every term on the top line contains at least one x .

Step 2: Factorise the denominator by extracting the common factor of x .

$$\frac{x(2 + xy - x)}{x(x + 3)}$$

This reveals a common factor on both the top and the bottom.

Step 3: Cancel the common factor of x to simplify fully.

$$\frac{2 + xy - x}{x + 3}$$

Divide the top and bottom by x .

Final answer: $\frac{2 + xy - x}{x + 3}$

Diagram.

$$\frac{\cancel{x}(2 + xy - x)}{\cancel{x}(x + 3)} \xrightarrow{\text{cancel } x} = \frac{2 + xy - x}{x + 3}$$

Cancellation diagram: factorise top and bottom, then strike the common factor.

Algebraic Fractions 45 (a3-045)

Difficulty: Foundation / Marks: 3

Simplify fully: $\frac{12cd-6c^2d+3c^3d^2}{12c^2de}$

Worked solution.

Step 1: Factorise the numerator by extracting the highest common factor of $3cd$.

$$\frac{3cd(4 - 2c + c^2d)}{12c^2de}$$

3 is the largest number that divides into 12, 6, and 3. Each term also contains at least a c and a d .

Step 2: Cancel the common factors of 3, c , and d from the numerator and denominator.

$$\frac{4 - 2c + c^2d}{4ce}$$

Dividing 12 by 3 leaves 4 on the denominator. Dividing c^2 by c leaves a single c , and the d cancels entirely. The e is unaffected.

Final answer: $\frac{4 - 2c + c^2d}{4ce}$

Diagram.

$$\frac{\cancel{3cd}(4 - 2c + c^2d)}{\cancel{3cd} \cdot 4ce} \xrightarrow{\text{cancel } 3cd} = \frac{4 - 2c + c^2d}{4ce}$$

Cancellation diagram: factorise top and bottom, then strike the common factor.

Algebraic Fractions 46 (a3-046)

Difficulty: Foundation / Marks: 3

Simplify fully: $\frac{x^2-9}{x^2+5x+6}$

Worked solution.

Factorise the numerator and denominator separately, then cancel.

$$\begin{aligned} \frac{x^2 - 9}{x^2 + 5x + 6} &= \frac{(x - 3)(x + 3)}{(x + 2)(x + 3)} \\ &= \frac{x - 3}{x + 2} \end{aligned}$$

The numerator is a difference of two squares. The denominator factorises as a quadratic. Cancel the common factor $(x + 3)$.

Final answer: $\frac{x - 3}{x + 2}$

Diagram.

$$\frac{(x - 3)\cancel{(x + 3)}}{(x + 2)\cancel{(x + 3)}} \xrightarrow{\text{cancel } (x + 3)} = \frac{x - 3}{x + 2}$$

Cancellation diagram: factorise top and bottom, then strike the common factor.

Algebraic Fractions 47 (a3-047)

Difficulty: Foundation / Marks: 2

Simplify fully: $\frac{ax+ay}{az}$

Worked solution.

Step 1: Factorise the top line by taking out the common factor of a.

$$\frac{a(x + y)}{az}$$

Both terms in the numerator share the variable a .

Step 2: Cancel the common factor of a from the numerator and denominator.

$$\frac{x + y}{z}$$

Dividing the top and bottom by a simplifies the fraction.

Final answer: $\frac{x+y}{z}$

Diagram.

$$\frac{\cancel{a}(x+y)}{\cancel{a}z} \xrightarrow{\text{cancel } a} = \frac{x+y}{z}$$

Cancellation diagram: factorise top and bottom, then strike the common factor.

Algebraic Fractions 48 (a3-048)

Difficulty: Foundation / Marks: 2

Express as a single fraction: $\frac{x}{3} + \frac{x}{4}$

Worked solution.

Step 1: Find a common denominator for 3 and 4, which is 12.

$$\frac{4x}{12} + \frac{3x}{12}$$

Multiply the top and bottom of the first fraction by 4, and the top and bottom of the second fraction by 3.

Step 2: Add the numerators together over the common denominator.

$$\frac{4x+3x}{12}$$

Combine the terms now that the denominators match.

Step 3: Simplify the numerator.

$$\frac{7x}{12}$$

Collect the like terms: $4x + 3x = 7x$.

Final answer: $\frac{7x}{12}$

Diagram.

$$\begin{array}{c} x \quad \quad x \\ \swarrow \quad \searrow \\ \text{---} + \text{---} \\ \uparrow \quad \downarrow \\ 3 \quad \quad 4 \end{array} = \frac{x \cdot 4 + x \cdot 3}{3 \cdot 4}$$

Cross-multiplication: each numerator multiplies the *other* denominator (curved arrows); the denominators multiply to give the common denominator.

Algebraic Fractions 49 (a3-049)

Difficulty: Foundation / Marks: 2

Simplify fully: $\frac{5x^2+10xy}{5x}$

Worked solution.

Step 1: Factorise the numerator by extracting the highest common factor, which is $5x$.

$$\frac{5x(x+2y)}{5x}$$

Look for the largest number and the highest power of the variables that divide exactly into both terms on the top.

Step 2: Cancel the common factor of $5x$ from the numerator and the denominator.

$$x+2y$$

Dividing the top and bottom by $5x$ removes the fraction entirely.

Final answer: $x+2y$

Diagram.

$$\frac{\cancel{5x}(x+2y)}{\cancel{5x}} \xrightarrow{\text{cancel } 5x} = x+2y$$

Cancellation diagram: factorise top and bottom, then strike the common factor.

Algebraic Fractions 50 (a3-050)

Difficulty: Foundation / Marks: 3

Express as a single fraction in its simplest form: $\frac{2}{x+3} - \frac{1}{x-1}$

Worked solution.

Step 1: Find a common denominator by multiplying the two linear denominators together.

$$\frac{2(x-1)}{(x+3)(x-1)} - \frac{1(x+3)}{(x+3)(x-1)}$$

Multiply the top and bottom of each fraction by the denominator of the other fraction.

Step 2: Combine the numerators over the single common denominator.

$$\frac{2(x-1) - 1(x+3)}{(x+3)(x-1)}$$

Subtract the second numerator from the first now that the denominators match.

Step 3: Expand the brackets on the numerator.

$$\frac{2x - 2 - x - 3}{(x+3)(x-1)}$$

Be careful with the negative sign when expanding the second bracket.

Step 4: Collect like terms to fully simplify the numerator.

$$\frac{x - 5}{(x+3)(x-1)}$$

$2x - x = x$ and $-2 - 3 = -5$.

Final answer: $\frac{x-5}{(x+3)(x-1)}$

Diagram.

$$\frac{\begin{array}{cc} 2 & 1 \\ \swarrow & \searrow \\ (x+3) & (x-1) \end{array}}{(x+3)(x-1)} = \frac{2 \cdot (x-1) - 1 \cdot (x+3)}{(x+3) \cdot (x-1)}$$

Cross-multiplication: each numerator multiplies the *other* denominator (curved arrows); the denominators multiply to give the common denominator.

Algebraic Fractions 51 (a3-051)

Difficulty: Foundation / Marks: 3

Express as a single fraction: $3 + \frac{2}{x-2}$

Worked solution.

Step 1: Rewrite the whole number 3 as a fraction with a denominator of $x-2$.

$$\frac{3(x-2)}{x-2} + \frac{2}{x-2}$$

Multiply the whole number by $\frac{x-2}{x-2}$ so it shares the same base as the fraction.

Step 2: Expand the numerator of the first fraction.

$$\frac{3x-6}{x-2} + \frac{2}{x-2}$$

Multiply the 3 into the bracket.

Step 3: Add the numerators together over the common denominator and simplify.

$$\frac{3x-6+2}{x-2}$$

Combine the constants: $-6+2=-4$.

Final answer: $\frac{3x-4}{x-2}$

Diagram.

$$3 + \frac{2}{x-2} \xrightarrow{\text{write } 3 = \frac{3(x-2)}{x-2}} \frac{3(x-2) + 2}{x-2} \xrightarrow{\text{simplify}} \frac{3x-4}{x-2}$$

Solution flow: each arrow shows one operation taking the expression to the next form.

Algebraic Fractions 52 (a3-052)

Difficulty: Foundation / Marks: 3

Simplify fully: $\frac{x^2-16}{4x+16}$

Worked solution.

Step 1: Factorise the numerator using the difference of two squares.

$$\frac{(x-4)(x+4)}{4x+16}$$

x^2 and 16 are both square terms separated by a subtraction sign.

Step 2: Factorise the denominator by extracting the common numerical factor of 4.

$$\frac{(x-4)(x+4)}{4(x+4)}$$

This reveals a common binomial bracket on the top and bottom.

Step 3: Cancel the common factor of $(x+4)$ from the numerator and denominator.

$$\frac{x-4}{4}$$

Divide the top and bottom by the shared bracket to leave the final simplified fraction.

Final answer: $\frac{x-4}{4}$

Diagram.

$$\frac{(x-4)\cancel{(x+4)}}{4\cancel{(x+4)}} \xrightarrow{\text{cancel } (x+4)} = \frac{x-4}{4}$$

Cancellation diagram: factorise top and bottom, then strike the common factor.
